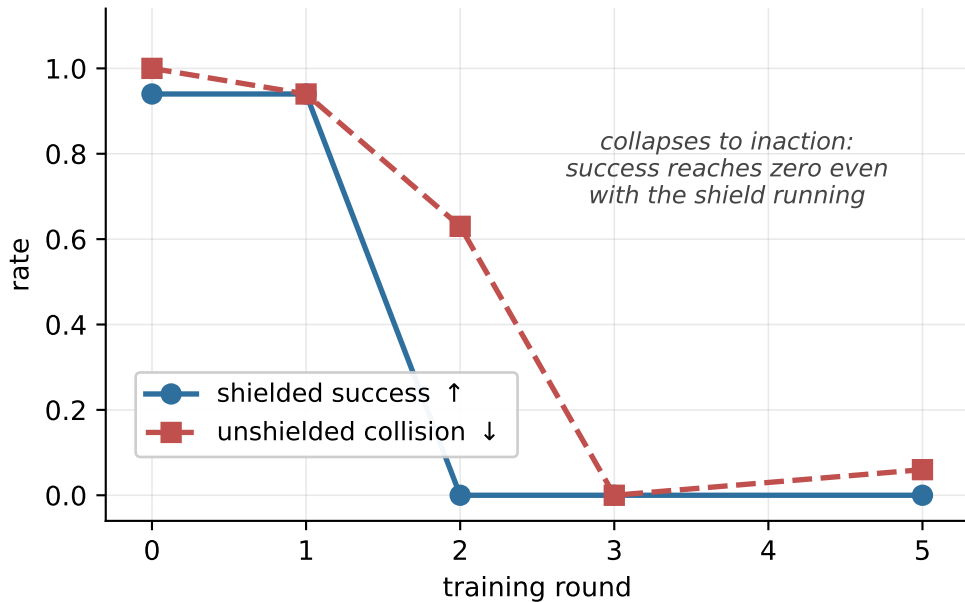


Scalar-reward RL: the failure bracketed from both sides

Exp 003 — collision enters the reward
(lr 5×10^{-5} , half rollouts unshielded)



Exp 004 — gentler gradient
(lr 2×10^{-5} , shield annealed $0.85 \rightarrow 0.40$)

